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Source: *Language*, Jan. - Mar., 1956, Vol. 32, No. 1 (Jan. - Mar., 1956), pp. 158-194

Published by: Linguistic Society of America

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A SEMANTIC ANALYSIS OF THE PAWNEE KINSHIP USAGE

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I. INTRODUCTION

1. LINGUISTICS, SEMANTICS, AND PRAGMATICS. The subject of this paper is not in strictest terms a linguistic one. 'Linguistics', in the narrow sense in which this term is currently used, is understood to concern itself solely with the formal properties of languages. That is to say, it is limited to the analysis of the properties of the signal systems or 'codes' themselves as inferred from the structure of messages, while it excludes on the one hand the primary nonlinguistic stimuli which prompt messages or are encoded in them, and on the other hand also the nonlinguistic responses which the messages may evoke. One need only remind oneself of Bloomfield's description of 'speaker's stimuli' and 'hearer's responses' for an illustration of the sorts of things that get excluded from linguistics.[1] These things which get excluded from linguistics proper, however, are to a large extent the concern and the subject matter of cultural anthropology, and they belong within the fuller study of language.

This division of the total context surrounding a linguistic event has been recognized by a number of writers. A well-known formulation is that of Charles W. Morris. For any sign system, linguistic or other, there are distinguished: (*a*) the properties of the signs and their systematic relations to each other, or 'syntactics'; (*b*) the relations of the signs to their areas of designation and their features of signification,[2] or 'semantics'; and (*c*) the relations of these to behavior in response to signs, or 'pragmatics'.[3] This scheme, applied to the study of language, would isolate the physical and the relational properties of linguistic signs as the subject matter of a special branch of study, while problems of meaning and of the social functions of language would constitute two additional fields of investigation. Precisely this departmentalization has been taking place in the last twenty-five years, and the first of these branches has crystallized as the science of 'linguistics'. The other two potential branches of the study of language have not been as systematically developed and have not been disentangled from each other in as clear-cut a manner as linguistics has been disentangled from them. They are dealt with in various manners according to the several tastes and interests of philosophers, lexicographers, philologists, general semanticists, anthropologists, and psychologists. They present problems partly of a different kind from those presented by linguistics, and partly of a similar kind though more complex.

The aim of this paper is to point out a relatively simple problem in semantics

[1] Leonard Bloomfield, *Language*, chapters 2 and 9 (New York, 1933).

[2] For the difference between 'designation' and 'signification' see §5.

[3] Charles W. Morris, *Foundations of the theory of signs* (International Encyclopedia of Unified Science, Vol. 1, No. 2; Chicago, 1938); *Signs, language and behavior* (New York, 1946).

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which can be analysed by means of techniques analogous to those already developed in linguistics, and to show some of the connections between pragmatics, semantics, and linguistics.

2. OVERT AND COVERT SEMANTIC CATEGORIES. Philosophers used to debate whether language reflects reality. Some still concern themselves with the problem of how accurately a language can be made to map reality. To the 19th-century European philosophers we owe also the view, which is currently popular among American anthropologists, that a language reflects the spirit or the culture of a people and that it influences their reactions to the world about them. It is asserted that both grammar and vocabulary hold clues to a people's world view. This, of course, amounts only to saying that language is meaningful, but that the meanings involved, and their classification, differ from one society to another; or that language is used in relating to the natural and social environment, but that different peoples experience differently and find varying significances in their environments. Of particular interest in this connection are the so-called obligatory categories in language—those features of meanings or of situations, real or imputed, of which the structure and usage of a language force recognition. It is with these that we shall be concerned in this paper. They are to be regarded as distinctive features of meaning.

The reflection, in language, of such distinctive features of the real or the cultural world, then, must be taken for granted. But there are various manners of reflection. In some cases it is revealed in the segmental structure of linguistic forms. In other cases it is not. For a simple illustration we may compare the following sets of Spanish and English forms:

tio	:	tia	uncle	:	aunt
hijo	:	hija	son	:	daughter
abuelo	:	abuela	grandfather	:	grandmother
hermano	:	hermana	brother	:	sister

The semantic (though not necessarily the pragmatic) relations obtaining between the terms of each pair are the same in both languages, viz. *male:female.* Yet in the one language this is reflected in the overt structure of the terms themselves, whereas in the other it is not. One might hastily conclude, then, that in Spanish this semantic relationship is something of interest to the linguist, but that in English it is not, since in the former language but not in the latter it makes it possible for him to segment the linguistic forms.

This, however, would be a premature conclusion. Systematic contrastive relations between linguistic forms are the subject matter of linguistics regardless of whether these contrasts have correlates in the segmental structure of the forms or consist simply in rules of selection and association. And as for native responses, speakers of English are as aware of the semantic relation involved in these pairs as are speakers of Spanish. The important fact is that the relation is recognized, in one way or another, in both languages. It compels us to take account of two equally important kinds of structure in language, one a structure of overtly represented categories and one a structure of covert categories.

Often there is a mixing of these two structural types in the recognition of a

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single semantic feature in forms of a language. Consider, for example, the following inflected forms from an Oneida verb paradigm:

katekhũ·níhe? *I am eating*	wa?katekhũ·ní· *I ate*
yakwatekhũ·níhe? *we are eating*	wa?akwatekhũ·ní· *we ate*
watekhũ·níhe? *she is eating*	ũtekhũ·ní· *she ate*
kũtekhũ·níhe? *they (fem.) are eating*	wa?kũtekhũ·ní· *they (fem.) ate*

A past-tense element *wa?-* is readily isolable in three of the forms of the second column. In these forms the past-tense feature finds an identity and a locus separate from those of the varying pronominal features. But in the third-person feminine singular form the past-tense and the pronominal features lose their separate identities and are merged in a single element *ũ-* which is incapable of further segmentation.

In suggesting a linguistic treatment for forms of this type Hockett considered the case of the French (spoken) form /o/, where

/o/:/ala/::/lə/:/la/.

The constituents of the second term are separable; the form can be segmented into a preposition and an article. In the first term, however, the constituents are inseparable and the form cannot be segmented. Hockett's proposal was to recognize dual morphemic membership for the unsegmentable term. It would be a morph having simultaneous membership in two different morphemes; its occurrence would be a token of the occurrence of both of these. He suggested the term 'portmanteau morph' for such forms.[4] According to this manner of description the prefixed element *ũ-* in the Oneida form for 'she ate' would be regarded as a portmanteau morph: it would belong, with *wa?-*, to the past-tense morpheme, and it would belong also, with *wa-* (and other morphs), to the third-person singular feminine morpheme;[5] its occurrence would signal at once the meanings of both of these morphemes.

A question soon arises how far this principle may be carried. Suppose we consider these pairs of English words:

baron	:	baroness	cat	:	kitten
duke	:	duchess	dog	:	pup
king	:	queen			

We may be willing to identify the constituents in *baroness* and *duchess*, and perhaps also in *kitten*, and to consider each of these as containing two morphemes. But shall we then recognize two morphemes in each of the 'portmanteau' morphs *queen* and *pup*? The semantic differences and the distributional differences between *queen* and *king* are parallel to those in the segmentable forms above them—as was the case also with the French /o/:/lə/ in comparison with /ala/:/la/, or with the Oneida *ũ-*:*wa-* in comparison with *wa?ka-*:*ka-*,

[4] Charles F. Hockett, Problems of morphemic analysis, *Lg.* 23.321–43, esp. 333 ff. (1947).

[5] Cf. my *Oneida verb morphology* 11–6, 35–45 (Yale University Publications in Anthropology 48; New Haven, 1953).

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wa?akwa-:*yakwa-*, and *wa?kũ-*:*kũ-*. We readily admit morphemic status for such semantic or distributional differences in some cases, but we balk at it in other cases. The concept of the portmanteau morph, with its multiple morphemic membership, is a clever device; but we are not consistent in using it.

Harris has called attention to a kind of case which has sometimes been treated in the 'portmanteau' manner. He suggested a different treatment, one in which such features are not elevated to morphemic status but are kept on a submorphemic level. The case that he discussed is that of paradigmatic sets in which the component categories do not always correspond to separately isolable morphs. It may be that they are nowhere isolable in the paradigm, or it may be that they can be individually localized in a few forms of the paradigm, but that taking the paradigm as a whole they cannot be consistently identified. We quote from his paper (italics ours):[6]

> The linguistic structure of an utterance is presumed to be fully stated by a list of the morphemes which constitute it, and by their order. The difference between two utterances is expressed by the difference in morphemic constituency between them. Frequently, however, we find that there is a set of morphemes in a language, *such that each morpheme in the set is identified by its contrast with all the others in the set.* Such morphemes are often arranged in paradigms, and the various crisscrossing relationships among the morphemes of the paradigm are often called categories. Thus there are such Latin morphemes as *-us*, *-um*, *-ī*, *-ōs* which are defined by their membership in a paradigm, and which are considered as expressing, within the paradigm, such categories as case and number.
>
> The presence of these categories is not a happy situation for structural linguistics, which is most useful if it can define everything in terms of some stock of elements (phonemes, morphemes) which are all on a par with each other. It is therefore of interest to note that the categories represented in a paradigm can be set up by means of the very methods which are used to set up the more traditional morphemes. ... One important difference between isolating the traditional morphemes and isolating these categories is that *the categories are not readily identifiable as consisting of any particular phonemes in the utterances.* ... We will therefore consider these morphemically analyzed categories COMPONENTS.

We may summarize our points as follows:

(1) Semantic features may be recognized in more than one way in a language. Some may be recognized overtly, with separate phonemic identities, while others may be recognized covertly, merged with other semantic features in various jointly and simultaneously shared phonemic identities.

(2) For a single semantic feature there is sometimes a mixing of the two manners of linguistic recognition: some features emerge, so to speak, at some points to find separate identity in the segmental structure of a language, but are submerged at other points, being identifiable only as possible contrasts between various already irreducible segments.

(3) There are two possible ways of treating these 'submerged' categories in linguistic description: (*a*) they may be given morphemic status, perhaps by

[6] Zellig S. Harris, Componential analysis of a Hebrew paradigm, *Lg.* 24.87 (1948).

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forcing the segmentation as far as it can be pushed and then by resorting to portmanteaus where segmentation is impossible; or (*b*) they may be given special submorphemic status as 'components'.

(4) The description of the componential structure of contrasting forms is an important part of linguistic analysis, whether or not the contrasts have any correlates in the segmental structure of forms. It should be noted here that in Harris's methodology the components, though labeled semantically, are defined distributionally. He does not deal directly with meanings, but with so-called 'linguistic meaning'—portions of possible linguistic contexts in which a form can occur, a privilege which it shares with certain other forms, though not with all, of the paradigm. This does not preclude the possibility of a direct semantic approach to componential structure, as we hope to demonstrate.

3. A PARADIGM-LIKE SET FROM LEXICON. Most componential analyses which have been made by linguists deal with paradigmatic sets of affixed morphemes.[7] Cases from lexicon, on the other hand, have bordered on the trivial, like the *kitten* and *pup* example and others referred to above. We may question, then, whether there is to be found a componential structure of covert features in lexicon also, or whether a lexicon may not be an unorganized patchwork with no interesting underlying principles, something which can only be catalogued, as in a dictionary, but which cannot be analyzed structurally. From what psychology tells us of the processes of learning, generalization, concept formation, and perception, and from what anthropology tells us of the nature of culture, we suspect that it is not a random patchwork, but that it may indeed be highly structured.

In the sections which follow, we shall be concerned with a lexical set whose componential structure is not so simple as to be trivial or uninteresting, nor yet so complex as to be insusceptible of analysis. The set is like a paradigm, inasmuch as the terms of the set can be defined as the products of components from a limited number of dimensions of difference. Yet it differs from the paradigms with which we are traditionally acquainted, inasmuch as no component is ever represented overtly in the segmental structure of the forms. The structure of the set is entirely one of covert categories.

Our analysis will be of the semantics of reference, rather than of linguistic distributions; the components are to be semantic features rather than distributional features. Contrary to the accepted fears concerning meaning, this proves to be a somewhat simpler task, and one admitting of greater precision, than the pursuit of distributional definitions. Our example comes from the work of anthropologists rather than from that of linguists. It is a kinship system.

[7] Cf. Roman Jakobson, Beitrag zur allgemeinen Kasuslehre, *Travaux du Cercle linguistique de Prague* 6.240–88 (1936); Thomas A. Sebeok, *Finnish and Hungarian case systems: Their form and function* (Acta Instituti Hungarici Universitatis Holmiensis, Series B, Linguistica 3; Stockholm, 1946); John Lotz, The semantic analysis of the nominal bases in Hungarian, *Travaux du Cercle linguistique de Copenhague* 5.185–197 (1949); William L. Wonderly, Semantic components in Kechua person morphemes, *Lg.* 28.366–76 (1952); Zellig S. Harris, *Lg.* 24.87–91 (1948).

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II. The Pawnee Kinship System

4. Data. Given below is a portion of the kinship vocabulary of the Republican Pawnee as we derive it from the data published by Lewis Henry Morgan.[8] For purposes of this paper we include only the consanguineal and the spouse terms. Definitions of meanings are given by listing as many kin types as are known to be included in each kin class.[9] By 'known' we mean either given directly by Morgan or inferable from reciprocal types or from sibling types given by him. In a very few instances rules of succession, when these are unambiguous, have been utilized also to fill in certain data. All types which are inferred in one of these ways are marked with an asterisk. Those without the asterisk are in Morgan's tables. As definitions, these listings are not exhaustive, for Morgan's schedule was not as complete as we should desire; yet they will suffice for an analysis.

Primary kin-types are abbreviated as follows: F = 'father', M = 'mother', B = 'brother', S = 'sister', s = 'son', d = 'daughter', H = 'husband', W = 'wife'. Secondary and higher-degree kin-types[10] are abbreviated with compound symbols: BW = 'brother's wife', MBd = 'mother's brother's daughter', FSs = 'father's sister's son', etc. All kin-types listed are assumed to be possible referents of the given kinship term for Egos of either sex, unless (*a*) a specific qualifier is added, such as 'for a male Ego', or (*b*) the kin-type is inherently restricted to Egos of only one sex, such as H and HM, which are applicable only by or in relation to a female.

Translation labels are also given for the Pawnee terms. These, of course, are not accurate either as definitions or as translations. They serve only to give us convenient English tags for the forms and their referents. In each case the translation label is the word which we would use in English, in our kinship usage, to refer to the nearest one of the kin-types in each Pawnee class. The extensions of each of these terms to additional kin-types, however, is very different in Pawnee from what it is in English. English terms when used as labels for Pawnee classes are written with double quotation marks, thus distinguishing them from the same words when used in their ordinary sense as labels for English classes. Thus "uncle" is not the same as 'uncle'.

[8] Lewis H. Morgan, *Systems of consanguinity and affinity of the human family* 195–9, 293–382 (Smithsonian Contributions to Knowledge, Vol. 17; Washington, 1871).

[9] An English example will illustrate the use of these terms. In our kinship system *father's brother*, *mother's brother*, *father's sister's husband*, and *mother's sister's husband* are four kin types belonging to the kin class *uncle*. For definitions and further discussion of these terms see §10.

[10] Primary kin types are those within the nuclear family. Since an individual normally becomes a member of (at least) two nuclear families—that of his parents and siblings, in which he finds himself as a child, and that of his spouse and offspring, which he establishes as an adult—it results that a primary relative of a primary relative is not necessarily also a primary relative. Some are; e.g. Ms = B, FW may = M. Some are not; e.g. MB and BW. The latter are secondary kin types. Similarly, a primary of a primary of a primary, which does not reduce to either a primary or a secondary, is a tertiary kin type; and so forth. Cf. George Peter Murdock, *Social structure* 92–7 (New York, 1949). The degree of a kin type is given, in our notation, by the number of letters used in its abbreviation.

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For convenience of reference in later sections, the types listed under each term are grouped by generation, with positive numbers for ascending generations, negative numbers for descending generations, and zero for Ego's own generation. Within each generation the types are further segregated by degrees of collaterality, semicolons being used for this purpose, while commas separate the types within any one degree of collaterality.

In some cases alternative classifications are indicated for a kin type. These alternatives are consistent with the structure of the system.

Republican Pawnee Kinship Terms

atípat "my grandfather"
- 2: FF, MF; FFB, *FMB (also *atías*[11]), *MFB, MMB (also *tiwátsiriks*?[12]); FFFBs, MMMBs
- 1: FSH; FFSs, *MFSs
- 0: FSdH; *FFSds
- −1: FFSdds

atíka "my grandmother"
- 2: FM, MM; FFS, *FMS (or *atíra*[13]), *MFS, MMS (also *atíra*?[14]); *FFFBd, FFFSd (also *atíra*?[15]), *MMMBd, MMMSd (also *atíra*?[16])
- 1: FFSd, *MFSd
- 0: FFSdd
- −1: FFSddd

atías "my father"
- 4: FFFF
- 3: *FFFFSs
- 2: *FMB (see *atípat*)
- 1: F, step-F; FB, MSH; FFBs, *FMSs; FFFBss; *FFFFBsss
- 0: FSs
- −1: FSds
- −2: FSdds
- −3: FSssss

atíra "my mother"
- 4: FFFM
- 3: MMM, MMMS, FFFS
- 2: *FMS, *MMS (see *atíka*)
- 1: M, step-M; MS, FS, FBW; MMSd, *MFBd, *FMSd, *FFBd; MMMSdd, *FFFBsd; MMMMSddd, FFFFBssd

[11] FMB is not given by Morgan. From Sss = *rakítiki* it can be reconstructed, by reciprocity, as *atípat*. From FSss = *irári/irátsti* it can be reconstructed, by reciprocity and succession, as *atías*.

[12] From FSds = *atías*, by reciprocity and succession.

[13] From *FMB = *atías*, by sibling relationship.

[14] From *MMB = *tiwátsiriks*, by sibling relationship. For the double signification of *atíra* see §7.4.

[15] From MBsss = *tíwat*, by reciprocity and sibling relationship.

[16] Parallel collateral extension from MMS. Cf. fn. 14.

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0: FSd, FSsW
−1: FSdd
−2: FSddd
−3: FSsssd

tiwátsiriks "my uncle"
3: FFF; FFFB, MMMB; *FFFFBs, *FFFMSs, *MMMMSs
2: *MMB (see *atípat*); *FFFSs
1: MB; *MFBs, *MMSs; *MMMSds

irátsti "my brother" (FOR A FEMALE EGO)
1: *FMBs
0: B, step-B; FBs, MSs; FFBss, MMSds; FFFBsss, MMMSdds
−1: FSss

itáxri "my sister" (FOR A MALE EGO)
1: *FMBd
0: S, step-S; FBd, MSd; FFBsd, MMSdd; FFFBssd, MMMSddd
−1: FSsd

irári "my brother" (FOR A MALE), "my sister" (FOR A FEMALE)
Included here are the types listed under *irátsti* and *itáxri* above, but with reversals of the sex of Ego.

píra?u "my child"

1. FOR A MALE EGO:
3: *FFFMBs, *FFFMBd
2: *MMMBs (also *atípat*), *MMMBd (also *atíka*)
1: *MMBs, *MMBd[17]
0: MBs, MBd
−1: s, d; Bs, Bd; *WSs, WSd; FBss, FBsd, MSss, MSsd; FFBsss, FFBssd; FFFBssss, FFFBsssd
−2: *Sss?, *Ssd? (see *raktíki*)
−3: MBssss, MBsssd
−4: ssss, sssd

2. FOR A FEMALE EGO:
Included are (*a*) all of the types listed above as *píra?u* for a male,

[17] From FSds = *atías* and FSdd = *atíra*, by reciprocity. Morgan gives MMBs as *tiwátsiriks*, which is inconsistent with the classification of the reciprocals FSds and FSdd. This is hardly to be interpreted as reflecting alternative principles of classification, but rather must be an isolated exception. Generally the child of an "uncle" is "child". The succession "uncle" > "uncle", in place of "uncle" > "child" is a known type of mutation in a Crow-type kinship system when rules of residence are changing from matrilocal to patrilocal. It is a shift from 'Crow' principles of classification to 'Omaha' principles of classification. In the Republican Pawnee data this must be an isolated instance of some informant's MMB having retained his parental residence after marriage, bringing his wife with him rather than going to live with his wife's family. That this must represent a single instance rather than a continuing new practice is indicated by the succession in the next and continuing generations where it goes "uncle" > "child" > "grandchild". As a result of this one exception the MMB agnatic line is displaced one classificatory generation upward in Morgan's data. We have deleted this divergent line from our data, and have filled it out by utilizing Morgan's data on reciprocals.

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with the exception of WSs and WSd; (*b*) also the following:
0: HMBs, HMBd
−1: HBs, HBd
and (*c*) also all of the types listed below as belong, for a male Ego, to the *tíwat* class.

tíwat "my nephew/niece" (ONLY FOR A MALE EGO)
−1: Ss, Sd; FBds, FBdd, MSds, MSdd; MMSdds, MMSddd
−2: *Sds, *Sdd (also *raktíki*); MBsss, MBssd
−3: sss, ssd; Bsss, Bssd, Sdds, Sddd[18]; FBssss. FBsssd, MSssss, *MSsssd, *MSddds, MSdddd

raktíki "my grandchild"
1: *MMMBss, *MMMBsd
0: *MMBss, *MMBsd
−1: MBss, MBsd, MBds, MBdd
−2: ss, sd, ds, dd[19]; Bss, Bsd, Bds, Bdd, Sss (or *píra?u?*[20]), Ssd (*píra?u?*), Sds, Sdd; FBsss, FBssd, FBsds, FBsdd, FBdss, FBdsd, FBdds, FBddd, MSsss, MSssd, MSsds, MSsdd, MSdss, MSdsd, MSdds, MSddd[21]; FFBssss, FFBsssd, FFBssds, FFBssdd, FFBsdss, FFBsdsd, FFBsdds, FFBsddd, MMSdsss, MMSdssd, MMSdsds, MMSdsdd, MMSddss, MMSddsd, MMSddds, MMSdddd[22]; FFFBssssss, MMMSdddddd

tatiráktaku "my wife"
1: MBW
0: W; WS, BW; FBsW, MSsW

tikútaku "my husband"
0: H; HB, SH; *HFBs, *HMSs
−1: *HSs

We can view a kinship system as effecting a segmentation of a portion of one's social environment. In the Pawnee system, we note that it is not the seg-

[18] The ambiguities in Morgan's data here, when he employs the terms 'great grandson' and 'great granddaughter' in defining kin types, are resolved in his special remarks on the Pawnee, op.cit. 196–7, and also by the data for the more exactly specified reciprocal types FFFB and MMMB, *tiwátsiriks*.

[19] The ambiguities here and in the following, which inhere in the terms 'grandson' and 'granddaughter', are resolved by the example of the specifically given MBss, MBds, MBsd, MBdd.

[20] See fn. 11 on the reciprocal type FMB.

[21] Morgan's ambiguous terms 'FB great-grandson' and 'FB great-granddaughter' are taken to include all eight possibilities, FB(s/d)(s/d)(s/d), because of the precedent furnished by (B/S)('grandson'/'granddaughter') = (B/S)(s/d)(s/d) = *raktíki*; cf. fn. 19. Similarly, 'MS great-grandson' and 'MS great-granddaughter' are taken as including MS(s/d)(s/d)(s/d).

[22] The ambiguous 'FFB great-great-grandson' and 'FFB great-great-granddaughter' are taken, on the same precedent, to include FFBs(s/d)(s/d)(s/d). Similarly, 'MMS great-great-grandson' and 'MMS great-great-granddaughter' are taken as MMSd (s/d)(s/d)(s/d). Cf. fns. 19 and 21.

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mentation which we would make of the same or of an analogous social environment. Note, by way of example, the following cases:

(1) The "uncle" term, *tiwátsiriks*. Of the four uncle types in first ascending generation in our usage (FB, MB, FSH, MSH), only one is included under the "uncle" term in Pawnee, viz. MB. The others are differently distributed: FB and MSH appear as "fathers", *atías*, while FSH is a "grandfather", *atípat*. On the other hand the "uncle" term in Pawnee includes what to us would be a great-grandfather, two great-great-uncles, and certain cousins once, twice, or thrice removed.

(2) The "nephew/niece" term, *tíwat*. Of secondary kin-types we note that this class includes only a man's sister's children, whereas in our usage a man's brother's children are also nephews and nieces. In Pawnee, however, we find these classified along with his own sons and daughters as "child", *píraʔu*. But besides his sister's children in the Pawnee nepotic class there are, as we would reckon them, certain great-grandchildren and remote cousin types. A woman, on the other hand, has no "nephews" or "nieces" in the Pawnee usage. All are her "children".

(3) Note that there is no "aunt" term. Of the types that we would consider aunts, FS, MS, and FBW are all classed as "mothers", *atíra*, in Pawnee, while MBW is known to a man simply as "wife", *tatiráktaku*.

(4) Note also the absence of a "cousin" term. We, in our system, are accustomed to class eight different kin-types as first cousins. In the Pawnee list above, we find four of these in the "brother" and "sister" classes, *irári*, *irátsti*, and *itáxri* (FBs, MSs, FBd, MSd). Two more are "father" and "mother", *atías* and *atíra* (FSs, FSd), while the remaining two are by Pawnee reckoning one's "children", *píraʔu* (MBs, MBd).

(5) Finally, by the rules of our society a man has not more than one woman whom he calls "wife" at any given time; but to a Pawnee a "wife", *tatiráktaku*, is not only his own wife, but his wife's sister, his brother's wife, and his mother's brother's wife, of whom two are sisters-in-law and one is an aunt in our reckoning.

5. DESIGNATUM, DENOTATUM, AND SIGNIFICATUM. There are two ways of defining a class: by naming the members of the class, or by stating the defining features of the class, that is to say, the necessary and sufficient conditions for membership in the class. These two ways are well known to us from phonemics. When we define a given phoneme by listing the allophonic types which belong to it, we are defining it by naming its members. When we define it as the class of phones sharing such-and-such distinctive phonetic features, we are defining it in the second way. In the case of our Pawnee kin classes we have begun with definitions of the first variety. These reveal one aspect of the structure of the system, namely the specific segmentation of the semantic field. They do not show, however, the underlying principles of organization. These will be shown only if we can proceed from the definitions-by-naming in such a way as to arrive at definitions in terms of distinctive semantic features, and further, if we can formulate the semantic structure of the whole set.

In the terminology which we take from C. W. Morris, we refer to particular

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referents of a term as DENOTATA; to the range of meaning, or class of denotata, as the DESIGNATUM of the term; and to the distinctive features of this class as the SIGNIFICATUM of the term.[23] For the Pawnee terms, then, any particular kin type is a denotatum of some term (e.g. MB is a denotatum of *tiwátsiriks*), while the class of all denotata of that term constitutes its designatum. Our goal is to discover the term's significatum.

6. DIMENSIONS OF KINSHIP DIFFERENCE. The data as they stand perhaps do not reveal at once any underlying structural principles. Closer inspection, however, aided by some advance knowledge of what to look for, reveals a number of features of patterning.

Our knowledge of what to look for is derived from the work of many students of diversely structured human societies. A now famous paper by Kroeber, published in 1909 under the title Classificatory systems of relationship,[24] gives us some clues. Kroeber noted: 'It is apparent that what we should try to deal with is not the hundreds or thousands of slightly varying relationships that are expressed or can be expressed by the various languages of man, but the principles or categories of relationship which underlie these. Eight such categories are discernible.' He then lists and describes the following kinds of differences between kinsmen, to which a kinship system may or may not accord recognition: (1) differences of generation; (2) the difference between lineal and collateral kinsmen, and, we should add, differences between kinsmen of varying degrees of collaterality;[25] (3) differences of age within one generation; (4) sex of kinsman; (5) sex of Ego; (6) sex of linking kinsman; (7) the difference between consanguineal and affinal relationships; and (8) the condition, living or deceased, of the linking kinsman. Although some of these have been found to require modification or further elaboration in specific instances, this list is still our guide, and we may use it in our examination of the Pawnee data.

As we look over the data, we see that generation is overridden in every kin class. Yet, as will be shown (§7), generation is basic to this system, if not so much as a dimension of absolute features, at least for the anchoring of relative features. This is quite different from its use in our English system, where it provides absolute features of considerable significance. To a limited extent it does this in Pawnee also, for one of the extensions of the "grandfather" class runs along generation lines.

[23] *Foundations* 3–9, 21–7; *SLB* 17–20 (cf. fn. 3). In the earlier work Morris used the term 'semantical rule' in place of 'significatum'.

[24] *Journal of the Royal Anthropological Institute* 39.77–84 (1909); reprinted in A. L. Kroeber, *The nature of culture* 175–81 (Chicago, 1952).

[25] 'Degree of collaterality' between a pair of consanguineal kinsmen is equal to the number of generations up a descent line of the higher-generation member of the pair, or of either if of same generation, up to their common ancestor. It is thus a reciprocal constant: the degree of collaterality of the first with respect to the second is equal to that of the second with respect to the first, regardless of which is taken as Ego. Degree of collaterality, thus defined, is the primary dimension of our English kinship system. Lineal kinsmen (ancestors and descendents), as well as oneself, are of zero degree collaterality; uncles and aunts, brothers and sisters, and nephews and nieces, as we use these kinship terms in English, are of first degree collaterality; while cousins are of second and higher degrees of collaterality.

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We note that degree of collaterality is nondistinctive in this system. This also is in marked contrast with our English system, where degree of collaterality furnishes the primary structural dimension. From the Pawnee data a negative structural principle emerges: parallel collateral kinsmen of the same sex and generation are everywhere merged. Thus, F, FB, FFBs, and FFFBss are all classed as "father", these ranging in degree of collaterality from zero to three.

As for the remaining kinds of kinship difference in Kroeber's list, we see that sex of kinsman is distinguished widely, though not everywhere; that sex of Ego makes a difference in two or three instances; and that sex of linking kinsman seems to be especially important. Relative age, and life condition of linking kinsman, on the other hand, are of no significance. We shall not concern ourselves here with the affinal contrast.

The dichotomy of consanguineal vs. affinal sometimes requires elaboration to accommodate distinctions which inhere in a people's kinship usage. Usage often implies distinctions between different kinds of affinity, and also between different kinds of consanguinity. As for the latter, the distinction may be that of *uterine* vs. *nonuterine*, or that of *agnatic* vs. *nonagnatic*. Some of the simpler Australian systems have these, and sex, as their only dimensions. Or the distinction may be of *parallel* vs. *nonparallel*.[26] These all relate to the distinction of sex of linking relative, but the question is one of formulation. For many systems of the Iroquois type, sex of the first and last in a chain of links is all that is important in this regard. For many other systems, however, special kinds of consanguinity need to be distinguished.

A people may behave as though one or another of these special kinds of consanguinity were more important than, or at least significantly different from, other kinds of consanguinity. Often the incest taboo applies with undiminished force to all parallel consanguineals regardless of how remote the connection, whereas for nonparallel kinsmen it may not apply, or its force may diminish rapidly with collateral distance. There are societies in which membership in a corporate kin-group is inherited in the uterine line, while some other status—social class, religious office, political prerogative, membership in another kind of kin-group, or an obligation to furnish a spouse—travels down the agnatic line. Not even every people's notion of blood relationship is the same. There are tribes who consider that an individual's body is derived from his mother in the female line, while his soul comes from his father in the male line.

For the Pawnee system we must make use of the notions of agnatic and of uterine consanguinity. These do not serve directly as dimensions of the system, but they are elementary distinctions necessary in the derivation of a more complex dimension of difference which is basic.

[26] For the definition of 'uterine' and 'agnatic' see §7. The term 'parallel' is sometimes understood as a cover term to include both of these: those to whom consanguinity can be traced entirely through connecting kinsmen of one sex. Sometimes, however, it applies only to kinsmen of the same generation, with the requirement that to be 'parallel' the parents through whom they are related must be merely consanguineal kinsmen (not necessarily themselves parallel) of the same sex. It is in this latter sense that it is applicable in the kinship systems of the Iroquois, the Cree, the Dakota, and other so-called Iroquois-type systems of North America.

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7. COMPONENTIAL DEFINITIONS. An anthropologist will recognize at once that this is a system of the so-called Crow type, a type widely found in many parts of the world. The principal stigmata of a Crow-type system are (1) the classification of the patrilateral first cross cousins [FSs, FSd] as "father" and "aunt", or the latter alternatively as "mother" or as "grandmother"; (2) the classification of the matrilateral first cross cousins [MBs, MBd] as "son" and "daughter", at least by a male Ego; and (3) the classification of both matrilateral and patrilateral first parallel cousins [MSs, MSd, FBs, FBd] as "brother" and "sister". The principal dimension of Crow-type systems is usually one which has the effect of opposing Ego's uterine kinsmen to his father's uterine kinsmen, and opposing these in turn to the father's father's and the mother's father's uterine kinsmen. This dimension cannot be understood or defined quite so simply, however, for types beyond these just named are included in each of the categories. As a secondary dimension, that of generation is usually given some recognition within the category of Ego's uterine kinsmen, but it is largely overridden within the other categories.

In the analysis of the consanguineal classes of the Pawnee system we may start with categories from the following five elementary kinship dimensions:[27]

(1) *Agnatic vs. nonagnatic consanguinity.* AGNATIC kinsmen, A, are consanguineals to whom relationship can be traced entirely through males; e.g. FBs, FFBss. NONAGNATIC kinsmen, $\bar{A}$, are those to whom relationship cannot be traced entirely through males.

(2) *Uterine vs. nonuterine consanguinity.* UTERINE kinsmen, U, are consanguineals to whom relationship can be traced entirely through females; e.g. MSs, MMSds. NONUTERINE kinsmen, $\bar{U}$, are those to whom relationship cannot be traced entirely through females. Agnatic vs. nonagnatic and uterine vs. nonuterine constitute separate dimensions, since the categories of the one are not mutually exclusive with those of the other. A kinsman may be both agnatically and uterinely related to Ego, for example one's brother, to whom one traces relationship through both father and mother. As we shall see (§7.10), there may be others.

(3) *Sex of kinsman.* This is reckoned partly in terms of the opposition MALE VS. FEMALE ($\sigma^1 : \varphi$), and partly in terms of the opposition SAME AS EGO'S VS. OPPOSITE TO EGO'S. The latter categories appear only in the definitions of sibling terms. For the most part we shall use the absolute categories rather than the relative categories.

(4) *Sex of Ego.* In the reckoning of reciprocal kin types, sex of Ego is the reciprocal of sex of kinsman. The categories are A MALE'S VS. A FEMALE'S ($\sigma^{1-1} : \varphi^{-1}$).

(5) *Generation.* This dimension presents the usual categories in ascending, zero, and descending generations: ... G^4, G^3, G^2, G^1, G^0, G^{-1}, G^{-2}, G^{-3}, G^{-4} ... These may be reckoned with respect to Ego or with respect to any other specified position within the system.

These are the elementary dimensions which we shall have recourse to. Con-

[27] These are 'elementary' only in a relative sense, being simpler than the derived dimensions which are used later. For a reduction of kinship dimensions to elementals see Joseph H. Greenberg, The logical analysis of kinship, *Philosophy of science* 16.58–64 (1949).

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siderable simplification of definitions can be achieved, however, if the first two of these are replaced by the following complex dimensions which incorporate generation distinctions along with consanguinity distinctions:

(1′) *Agnatic generation.* This dimension isolates generation categories as subclasses within the class of agnatic kinsmen: ... A^4, A^3, A^2, A^1, A^0, A^{-1}, A^{-2}, A^{-3}, A^{-4} ..., where $A^n = A \cdot G^n$, i.e. agnatic kinsmen of the nth generation. These may be reckoned with respect to Ego or with respect to any other specified position within the system. The totality of this dimension is simply A, the class of all agnatic kinsmen of any specified individual. All of the degrees from A^4 to A^{-4} will be needed in the analysis of the Pawnee data.

(2′) *Uterine generation.* This dimension isolates generation categories as subclasses within the class of uterine kinsmen. These may be reckoned with respect to Ego or with respect to any other specified position within the system. The totality of this dimension is U, the class of all uterine kinsmen of any specified individual. For the analysis of the Pawnee data we shall need from this dimension only the categories U^+, U^0, U^-, together with their totality U; where $U^+ = U \cdot G^+$, uterine kinsmen of all ascending generations, $U^0 = U \cdot G^0$, uterine kinsmen of one's own generation, and $U^- = U \cdot G^-$, uterine kinsmen of all descending generations.

A still greater simplification of the definitions of the Pawnee kin classes is achieved finally by introducing a further derived dimension, that of *agnatic rank*, **A**. It is assumed that the greater the simplicity of the definitions, the closer is the approximation to the dimensions and categories proper to the system, however complex these may appear when expressed in terms of the elementary distinctions of kinship axiomatics. The dimension of agnatic rank is not employed at the outset, however, but is reserved until that point in the development where the concept arises naturally in the formulations of the kin classes. The definition is found in §7.8.

We may now consider the data and put our categories to use. We have chosen to move inductively from the data, rather than to present the reader with a fait accompli. This has seemed the preferable procedure in view of the fact that at every turn the data present problems which require special consideration. In §§7.1–7 we carry the formulations only as far as the immediate data for the various classes require. In §7.8 we relate these to each other and to the structure of the system as a whole, and reformulate the class definitions accordingly.

7.1. Within the "uncle" class, *tiwátsiriks*, there is a group of kin-types [MB, *MMSs, *MMMSds; *MMB; MMMB, *MMMMSs], to all of which Ego can trace relationship entirely through females. They are thus his uterine kinsmen. Semicolons in the listing segregate them by generation. The first three are U^1; the next is U^2; the last two are U^3. Generalizing from these, and attending to contrasts with all other classes, we may characterize this group of "uncles" as *males* who are Ego's *uterine kinsmen of ascending generations*: $♂ \cdot U^+$. This formulation furnishes a basis for prediction or extrapolation to certain other kin-types not included in Morgan's data. We might predict that all of an individual's male uterine kinsmen of ascending generations could be called "uncle". While we do not have access to data which would verify this for the

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Pawnee, it is entirely consonant with what we know of Crow-type kinship systems in general, on which there is an ample literature.

This formulation does not include all of the "uncle" types, however. MFBs is not a uterine type with respect to Ego. There remains the possibility, however, that due to some restriction on marriage practices, such an individual might be doubly related to Ego and thus be a uterine kinsman to him after all, by virtue of his other line of connection. We know that it was a common practice among the Pawnee for two or more men who were brothers to take as wives women who were sisters. Thus, a MFBs would very frequently be simultaneously a MMSs also, and hence a uterine kinsman to Ego. Yet, this was apparently not always the case, for we are told that brothers did sometimes marry women who were not sisters. Whether the latter would then be classificatory "sisters" (i.e. first, second, or more distant parallel cousins) we are not told for certain, though the implication is that they very often were. In any case, the classificatory feature involved here is a general rule in systems of this type, whether or not a marriage restriction of this sort is involved. Hence, for the sake of general applicability we may modify the formulation of the preceding paragraph so as to include not only Ego's male uterine kinsmen of ascending generations, but also the same-generation agnatic parallels to any of these. We may now say that "uncles" are *males* who are *same-generation agnatic kinsmen of any of Ego's ascending-generation uterine kinsmen*: $♂ \cdot U^{+}A^{0}$. The class thus defined[28] includes as a subclass the one previously defined, for any individual is by definition a same-generation agnatic kinsman (A^{0}) to himself. While this extension was made to provide for the inclusion of a single kin-type [MFBs] which we happen to have in our data, it obviously provides for (or predicts) the inclusion of many more, e.g. MMFBs, MFFBss, etc. These predictions are in harmony with what is known of Crow-type systems in general, and are not contradicted by any of the data at our disposal.

To show how the formula accommodates the data, we may tabulate as follows:

	U^{+}	A^{0}
	MB	*self*
or	M	B
	MMSs	*self*
	M	FBs

[28] Note that the formula juxtaposes symbols in two manners: with intervening dot, and without. Juxtaposition without the dot indicates a dependency relationship analogous to our genitive construction. The meaning of A^{0} cannot be defined absolutely as can the meaning of ♂. It is relative to a chosen reference point. If no other reference point is prefixed, it is understood to be Ego. Thus, A^{0} is to be read, *Ego's agnatic kinsman of the same generation as he.* Some other reference point may be stated, however, as in $U^{+}A^{0}$. This is to be read as *Ego's ascending-generation uterine kinsmen's agnatic kinsmen of same generation as they.*

Juxtaposition with intervening dot indicates the Boolean class product. $X \cdot Y$ is the class of things which are both X and Y, i.e. the intersection of the two given classes. Dot products are commutative. That is, $X \cdot Y = Y \cdot X$. Simple juxtapositions are not comnutative. $XY \neq YX$.

We shall have occasion also to use the Boolean sum. $X + Y$ is the class of things which are either X or Y. Cf. George Boole, *The mathematical analysis of logic* (Cambridge, 1847).

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etc. (The reader may complete the testing and may make predictions beyond the given data.)

There are still other "uncle" types in Pawnee [FFF, FFFB, *FFFFBs, *FFFMSs; *FFFSs] whose inclusion in this class is unusual for Crow-type systems. All of these hinge on the FFF. Three of them [FFF, FFFB, FFFFBs] can be described as A^3, i.e. as Ego's agnatic kinsmen of the third ascending generation. All of them can be brought under the formulation A^3U, thus:

	A^3	U
	FFF	*self*
	FFFB	*self*
or	FFF	B
	FFF	MSs

etc.

If the Pawnee included these types in the same class with the preceding types, we must assume that there was some feature of similarity which all of these types shared. Now, this feature does not inhere in the types themselves. It could have been there only if some Pawnee practice resulted in the identification of A^3U with U, i.e. of the father's father's father's uterine kin group with Ego's own uterine kin group. Recorded Pawnee ethnography does not provide us the answer. We can speculate, but this will be reserved for a later section. At the moment we are concerned with semantic analysis, not with pragmatics. Here we shall simply posit $A^3U = U$. This is not an isolated involution of the system, for we shall discover that Pawnee classifications imply the further identifications $UA^{-3} = U$, $A^4U = A^1U = UA^{-2}$, $A^2U = UA^{-1} = UA^{-4}$, etc. Morgan describes with some explicitness how one calls one's father "father", and his father in turn "grandfather", and the latter's father "uncle" (same term as for the mother's brother), and his father "father" again, and so on indefinitely in a three-generation cycle. Similarly he describes a three-generation cycle in descending generations, going "son", "grandson", "nephew" (same as sister's son), "son", etc. indefinitely.

Admitting, then, the equation $A^3U = U$, we are able to subsume all of the given "uncle" types under our previous formula, $♂ \cdot U^+A^0$. If, as seems indicated from ethnography, marriages of brothers or classificatory "brothers" to sisters or classificatory "sisters" was quite general, the formula virtually reduces to $♂ \cdot U^+$, i.e. male uterine kinsmen of ascending generation. Our more general expression, however, provides for the inclusion also of any same-generation agnatic parallels to these, who may not have been also uterinely connected to Ego.

7.2. The "nephew/niece" class, *tíwat*, is the reciprocal of the "uncle" class. Anyone who calls Ego *tiwátsiriks* will be called *tíwat* by him in return. The definition of the "nephew/niece" class can then be derived by taking the reciprocal of that for the "uncle" class. Thus: "nephew/niece" = "uncle"$^{-1}$ = $(♂ \cdot U^+A^0)^{-1}$ = $♂^{-1} \cdot A^0U^-$, i.e. *a man's same-generation agnatic kinsmen's descending-generation uterine kinsmen.*[29]

[29] Nondot juxtapositions, being noncommutative, require inversion of order when taking reciprocals. Dot products, being commutative, may be written in any order. ♂ = *male*. $♂^{-1}$ = *a male's*. Generation reciprocals change sign. Zero-generation is self-reciprocal.

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The reciprocal of SEX OF KINSMAN is SEX OF EGO. The "uncle" class is specific as to sex, but is unspecific as to sex of Ego. ("Uncles" are male; both males and females call the same kin-types "uncle"; there is no corresponding "aunt" class.) The "nephew/niece" class, conversely, is specific as to sex of Ego, but is unspecific as to sex of kinsmen. (Only a man has "nephews or nieces"; the class includes both sexes.)

In the above formulation it must be assumed that $UA^{-3} = A^0$, which is the reciprocal of the assumption posited in §7.1.

The formula may be tested against the data in the following manner:

		A^0	U^-
(*a*)		*own*	Ss/d
	or	S	s/d
		FBd	s/d
		own	MSds/d
		own	Sdds/d

		U	A^{-3}	U^-
(*b*)		*own*	sss/d	*self*
		own	Bsss/d	*self*
	or	B	sss/d	*self*
		MB	sss/d	*self*
		own	FBssss/d	*self*

etc. (Ss/d is to be read *sister's son or daughter*, and similarly in all instances with the slant line.)

7.3. The "father" class, *atías*, can be defined as *males* who are *uterine kinsmen of Ego's first-ascending-generation agnatic kinsmen*, i.e. $♂ \cdot A^1U$, provided we posit the identity of A^4 with A^1 (cf. §7.1) and that of UA^{-3} with U (cf. §7.2). The data may be matched against the formula as follows. For convenience we use the form $(A^3)A^1U(A^{-3})$, in which A^4 is factored into A^3A^1, and where the parenthesized factors, because of our identifications, may be omitted in any instance.

(A^3)	A^1	U	(A^{-3})
—	F	*self*	—
—	FFBs	*self*	—
—	F	MSs	—
—	FS	s	—
—	FS	ds	—
—	FS	s	sss
FFF	F	*self*	—

The remaining types may be filled in in similar manner.

Two affinal types [step-F, MSH] are included in the "father" class also. Given the practice of the levirate (see §8) and of frequent marriages of brothers to women who are each others' sisters, the probability is high that a stepfather and a mother's sister's husband would be a brother or parallel cousin to one's father, and hence a "father" to begin with. Hence the identification of these

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statuses and the ascription of "father" status to a stepfather and a MSH in all cases.

7.4. The "mother" class, *atíra*, cannot be given a unitary definition in terms of the components of the system. It is a composite class, paralleling on the one hand the "uncle" class, and on the other hand the "father" class. It consists of the sisters of all of those called by either of these two terms. The definition must be expressed as the sum of the two classes $♀ \cdot UA^0$ and $♀ \cdot A^1U$, where the same three-generation cyclical identifications are posited as in §7.1 and §7.3.

Unitary definitions are a desideratum. If a number of denotata are classified together, we usually assume that they must have some features of similarity which may constitute the defining features of the class. Otherwise it is not really one class, but two. Such is the case here. It may be said to be a case of 'polysemy', two different meanings for the same word. It is interesting to note that the Pawnee consider their "mother" term in precisely this light. Informants have been apologetic in regard to their having no separate term in their language for the second of the classes defined above (the "aunt" class of Crow-type systems: FS, FSd, etc.). They say that they call these "mother" because of the great respect which they have for them, and there is no other term so expressive and affect-laden as *atíra*.[30] There is a similar emotional basis for another extension of this term: its application to corn, a sacred food.

7.5. The "child" class, *píra?u*, for a male Ego, is the reciprocal of the "father" class: $(♂ \cdot A^1U)^{-1} = ♂^{-1} \cdot UA^{-1}$. That is, for a male Ego a "child" is any *first-descending-generation agnatic kinsman of any uterine kinsman.* This formulation assumes that $A^{-4} = A^1$, and that $A^3U = U$, the reciprocal assumptions to those of §7.3. The data may be matched against the formula in its form $(A^3)UA^{-1}(A^{-3})$, where A^{-4} is factored into $A^{-1}A^{-3}$, and where the parenthesized factors may be omitted in any instance.

(A^3)	U	A^{-1}	(A^{-3})
FFF	MB	s/d	—
—	MMB	s/d	—
—	*own*	FBss/d	—
—	MB	s	sss/d

The remaining types may be checked against the formula in similar manner.

The "child" class, for a female Ego, is the reciprocal of the "mother" class. Since the "mother" class parallels both the "uncle" and the "father" classes, its reciprocal includes all the kin types which are included, for a male Ego, in the "child" and in the "nephew/niece" classes. A woman has no "nephews" and "nieces". The definition of this composite class can be expressed as the sum of those of the "child" and "nephew/niece" classes given above, changing only the sex-of-Ego sign from $♂^{-1}$ to $♀^{-1}$.

7.6. The "brother" class, *irári* to a male Ego, *irátsti* to a female Ego, is seen to include one's male uterine kinsmen of his own generation [B, MSs, MMSds,

[30] Gene Weltfish, oral communication.

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MMMSdds] and also one's male agnatic kinsmen of the same generation [B, FBs, FFBss, FFFBsss]. The former are $♂ \cdot U^0$; the latter are $♂ \cdot A^0$. These can be brought together in an expression involving a class sum: $♂ \cdot (U^0 + A^0)$.

But there are other kin types within the "brother" class [FMBs, FSss] which are neither uterine nor agnatic types.[31] Nor are they even of Ego's own generation. They are, therefore, disqualified on every count for being U^0 or A^0. It is of interest to note, however, that they are the offspring of men [FMB, FSs] who are A^1U, i.e. "fathers". This is a characteristic which they share with the A^0 group noted above. We may combine these two groups, then, into one, which can be characterized as $♂ \cdot A^1UA^{-1}$. The A^0 group is just a special case within this more general case: the case where the U component is restricted to U^0. All of the "brother" types may now be brought under the revised formulation: $♂ \cdot (U^0 + A^1UA^{-1})$.

Some of the members of the U^0 group also may be brought under the A^1UA^{-1} characterization. Brothers [B] automatically belong to both groups. Even half-brothers by different fathers would also automatically belong to both groups because of the Pawnee practice of the levirate.[32] The uterine parallel cousins [MSs, MMSds, MMMSdds] would frequently have agnatic connections also, because of the quite common practice of brothers marrying women who were sisters. But this would not necessarily always be the case. Hence we are still left with a composite class, as formulated in the expression at the end of the preceding paragraph, even though the two classes which enter into the sum are seen to overlap.

Since U is a subclass of A^0UA^0, and since A^0UA^0 and A^1UA^{-1} can be brought under the single formulation A^nUA^{-n}, this last expression may be of significance in connection with the definition of the "brother" class. It is seen to include both parts of the "brother" class. But it is seen also to include the "uncle" and "nephew" classes as previously defined. These three classes are significantly similar in many respects. Not only can they be brought together under a single cover definition, but they are functional equivalents in a number of contexts. For example, the children of "brothers", of "uncles", and of "nephews", as well as a man's own children, are all his "children". This is characteristic of Crow-type kinship systems in general.[33] If we index the uterine dimension in the above expression, we can define the "brother" class as $♂ \cdot A^nU^mA^{-n}$, where, if $n = 0$, then $m = 0$, but where if $n \neq 0$, then m is unrestricted. This is the residue of the total class after the "uncle" subclass ($n = 0$, m positive) and the "nephew" subclass ($n = 0$, m negative) have been taken out. The relation of

[31] Neither could kinsmen of such types be uterinely or agnatically connected to Ego by virtue of some other line of connection. For this to be the case, there would have to have been a practice of first-cross-cousin marriage, which however is exceedingly rare for peoples with Crow-type kinship systems, and which for the Pawnee would have been incestuous.

[32] 'Levirate' is the practice whereby, when a man dies and leaves a widow, his brother (usually a younger brother) is obliged to take the widow to wife.

[33] Some of these, which make use of the relative-age distinction, simply extend their 'elder brother' term to the "uncle" types and their 'younger brother' term to the "nephew" types.

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this formulation to those of the other kin classes and to the structure of the system will become clearer with the portrayal of the total structure (§7.8).

7.7. The "sister" class, *itáxri* to a male Ego, *irári* to a female Ego, parallels the "brother" class and can be given a similar definition, changing only the sex sign. These definitions suffice when the sex of Ego is specified in any given case. In the general case, however, sex must be specified as follows:

	SEX OF EGO	SEX OF KINSMAN
irári	either	same as Ego's
irátsti	female	opposite
itáxri	male	opposite

7.8. With formulations which cover the data for the "uncle", "nephew/niece", "father", "mother", "child", "brother", and "sister" classes, we are now in a position to consider the structure of the system as a whole. The composition of the "grandfather", "grandmother" and "grandchild" classes can then be described better in relation to the rest of the system.

It has already been noted (§7.6) that the combined "brother", "uncle", and "nephew" classes can be brought under a single formulation. If we include also the female siblings to these, i.e. the "sister" class, the first or basic "mother" class, and the females in the "nephew/niece" class, we may bring all of these under the single formulation A^nUA^{-n}. To this joint class are opposed two further classes as follows.

In consideration of the data given for the "father" class, we arrived at a formulation of that class as $♂ \cdot A^1U$. This was adequate to cover the given data. Yet it is generally true of Crow-type systems that the "father" term is applied not only to one's father, but to anyone whom his father calls "brother". This is also said to be true for the Pawnee. The class would include, then, certain other types not given in our basic data and not covered by the above formulation. Examples of such "fathers" would be FFMBs and FFSss, for the father's FMBs and his FSss are both his "brothers" (cf. 7.6). These are $♂ \cdot A^2UA^{-1}$. The general formulation which would include all such cases is $♂ \cdot A^{n+1}UA^{-n}$.

The "child" class for a male Ego is the reciprocal of the "father" class. Our previous formulation, $♂^{-1} \cdot UA^{-1}$ (§7.4) was adequate for the data given. If we broaden the class in line with the broadening of the "father" class just noted, and if we include the second "child" class for a female Ego, we have the more general form A^nUA^{-n-1}.

In all of the above formulations, n is restricted to positive values or zero. Allowing negative values for n would introduce extraneous kin types into each of the classes. E.g. if it were allowed that $n = -1$, then MF and SH (= SsF) would be introduced into the "father" class. But the Pawnee classify these as "grandfather" (*atípat*) and "brother-in-law/son-in-law" (*kustáwextsu*), respectively. Similarly, WF (= sMF) would be introduced into the "brother-uncle-nephew" joint class, whereas by Pawnee terminology he is a "father-in-law" (*kúrahus*). Similarly also, WB (= sMB) would be introduced into the "child"

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class, but the Pawnee treat him as a "brother-in-law" (*takútskurus*). The essence of this restriction is that these formulations hold for reckoning 'upwards, over, and down', or 'upwards and over', or 'over and down', or simply 'over' (cf. Figure 1); but not for reckoning 'downwards, over, and up', or 'downwards and over', or 'over and up'. This is an understandable restriction, for the ascription of consanguineal kinship status on the basis of lower-generation connecting links is not usual in any society.

The major dimension in this kinship system is that which opposes these three segments of an individual's kindred to each other and to the residue of the kindred which is not contained in these. We have:

$A^{n+1}UA^{-n}$	"fathers" and the sisters of these
$A^{n}UA^{-n}$	"brothers", "uncles", and "nephews" and the sisters of these
$A^{n}UA^{-n-1}$	"children"[34]

Each of these major classes consists of a group of kin types which can be reached from Ego's position in a genealogical chart by ascending some number of agnatic generations (any low positive number or zero), and then descending a determined number of agnatic generations, not just from the point reached in the ascent, but from *any point on the uterine line which intersects it.* Figure 1 illustrates this principle as it applies to the $A^{n}UA^{-n}$ or "brother-uncle-nephew" class.

Since such a manner of reckoning, which measures distance in the agnatic dimension but ignores it (for the most part) in the uterine dimension, is basic to the Crow-type system, we may recognize it as such and define features of AGNATIC RANK, **A**, as follows: $\mathbf{A}^{1} = A^{n+1}UA^{-n}$, $\mathbf{A}^{0} = A^{n}UA^{-n}$, $\mathbf{A}^{-1} = A^{n}UA^{-n-1}$.

We have started from the elementary concepts of agnatic consanguinity, uterine consanguinity, and generation; and following the demands of the data, we have arrived at the above derived concepts of agnatic rank. If now, these be taken as constituting the basic dimension, some simplification can be achieved in the forms of the kin-class definitions. We shall have:

(1) The "father" class: $♂\cdot\mathbf{A}^{1}$; i.e. males of agnatic rank *one*.

(2) The joint "brother-uncle-nephew" class: $♂\cdot\mathbf{A}^{0}$; i.e. males of agnatic rank *zero*.

(3) The "uncle" class: $♂\cdot\mathbf{A}^{0}\cdot U^{+}$; i.e. males of agnatic rank *zero*, who are uterine kinsmen of ascending generations.

(4) The "mother" class: $♀\cdot(\mathbf{A}^{0}\cdot U^{+} + \mathbf{A}^{1})$; i.e. females of agnatic rank *zero* who are uterine kinswomen of ascending generations, together with females of agnatic rank *one*.

(5) The "brother" class: $♂\cdot\mathbf{A}^{0}\cdot(U^{0} + \bar{U})$; i.e. males of agnatic rank *zero*, who are either uterine kinsmen of zero generation or nonuterine kinsmen of any generation.

(6) The "sister" class: $♀\cdot\mathbf{A}^{0}\cdot(U^{0} + \bar{U})$; i.e. females of agnatic rank *zero*, who are either uterine kinswomen of zero generation or nonuterine kinswomen of any generation.

[34] This is the entire "child" class to a male. To a female it is only half of the "child" class. Cf. §7.5.

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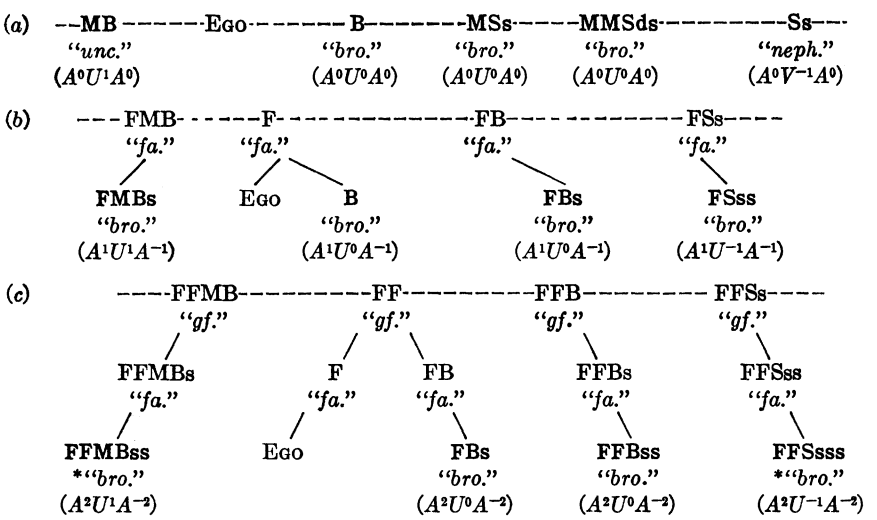


FIGURE 1

Some "Brother", "Uncle", and "Nephew" Types, $A^nU^mA^{-n}$, and their Connections to Ego

Solid lines represent agnatic connections; broken lines represent uterine connections. (*a*) $n = 0$; (*b*) $n = 1$; (*c*) $n = 2$. Where $n = 0$, the value of m (positive, negative, or zero) is distinctive. Where $n \neq 0$, the value of m is nondistinctive. The generation of any of these "brothers" is equal to the sum of the numerical indices on A and U. Those illustrated fall within the first ascending, the first descending, and Ego's own generation. Note that certain types in Ego's generation may be reckoned in either of two manners: B, as the son of Ego's mother, is $A^0U^0A^0$; and as the son of Ego's father, is $A^1U^0A^{-1}$. Similarly FBs is $A^1U^0A^{-1}$ or $A^2U^0A^{-2}$. The two starred listings *"bro." are extrapolations or predictions from the pattern. All other "bro." listings are supported in Morgan's data.

(7) The "nephew/niece" class: $♂^{-1}\cdot\mathbf{A}^0\cdot U^-$; i.e. only with respect to a male Ego, persons of agnatic rank *zero*, who are uterine kinsmen of descending generations.

(8) The "child" class: $♂^{-1}\cdot\mathbf{A}^{-1}$; $♀^{-1}\cdot(\mathbf{A}^0\cdot U^- + \mathbf{A}^{-1})$; i.e. with respect to a male Ego, persons of agnatic rank *minus one*; and with respect to a female Ego, persons of agnatic rank *zero* who are uterine kinsmen of descending generation, together with persons of agnatic rank *minus one*.

7.9. The "grandfather" class, *atípat*, contains a group of kin types [FF, FFB, FFFBs, FFSs, FFSds, FFSdds] which fall within the characterization $♂\cdot\mathbf{A}^2$. It contains another group [MF, MFB, MFSs] who are "fathers" of one's own mother and mother's brother, and who are thus $♂\cdot U^1\mathbf{A}^1$. Finally, the "grandfather" class cuts across the "father", the "uncle", and the "child" classes at the second ascending generation level to include, respectively, the FMB, the MMB, and the MMMBs. These, together with some of those already named, constitute a group [FF, FFB, FFFBs, MF, MFB, FMB, MMB, MMMBs] which is simply $♂\cdot G^2$.

The FF would seem to be the pivotal member of the "grandfather" class.

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The $\mathbf{A}^2$ extension represents one line of generalization from this type, a generalization along a uterine line. The G^2 extension, intersecting other classes, represents another line of generalization from the same type, this time along a generation line. The G^2 generalization brings in, among others, the MF, a type not meeting the conditions for membership in any other kin class, and thus, part of a residue. With its classification as "grandfather" it becomes a secondary pivotal member of this class and brings in with it the $U^1\mathbf{A}^1$ extension, a generalization along another uterine line.

The "grandfather" class is thus a composite class: $♂\cdot(\mathbf{A}^2 + G^2 + U^1\mathbf{A}^1)$; i.e. males who are of agnatic rank *two*, or of second ascending generation, or of agnatic rank *one* with respect to first-ascending-generation uterine kinsmen.

Besides the consanguineal types which have been categorized here, there are two affinal types [FSH, FSdH] which are listed as belonging to the "grandfather" class. These come in by virtue of ascribed agnatic rank $\mathbf{A}^2$: they are fathers of "fathers" [FSs, FSds]. As in other cases, a male kinsman's agnatic rank is one higher than that of his spouses. Thus, F is $\mathbf{A}^1$ while M is $\mathbf{A}^0$; FF is $\mathbf{A}^2$ while FM is $\mathbf{A}^1$. So also, FSH and FSdH are assigned as $\mathbf{A}^2$, inasmuch as FS and FSd are $\mathbf{A}^1$.

The "grandmother" class, *atíka*, is the female counterpart or sister class to the "grandfather" class. The formulation is the same, except for the change of the sex sign. In its G^2 extension it intersects the "mother" and the "child" classes, taking in the MM, MMS, MMMSd, the FM, and the MMMBd.

The "grandchild" class, *raktíki*, is the reciprocal of the "grandfather" and "grandmother" classes: $\mathbf{A}^{-2} + G^{-2} + \mathbf{A}^{-1}U^{-1}$; i.e. persons who are of agnatic rank *minus two*, or persons who are of second descending generation, or persons who are first-descending-generation uterine kinsmen to those of agnatic rank *minus one*. In its G^{-2} extension it intersects the "child", the "nephew/niece", and the "father" and "mother" classes (the reciprocals of those intersected by the "grandfather" and "grandmother" classes), taking in such types as ds, dd, Sds, Sdd, and, we might expect (though the data do not substantiate the prediction), the FSdds and FSddd.

The existence of class intersections, that is, of alternative classifications for certain kin types, would seem to be a blemish in the system. In any case it is an inconsistency which begs for an explanation. We cannot answer the question definitely, but we can put forward two hypotheses. The one is that the choice of label is contextually determined; e.g. that less formal situational contexts may call for the more familiar "grandparental" terms, whereas in other situations the same kin types might be classified differently, and more properly, on the basis of their other features. The other hypothesis is that the data which Morgan presents were collected from more than one informant and that we have represented here differences in semantic dialect. It is of particular interest in this connection to note that the classification of lower-generation kin types is more consistent with the basic pattern of the system, while the classification of the higher-generation reciprocal types in a number of instances follows the inconsistent G^2 generalization, or is deviant in some other way. Thus, FSdds and FSddd are 'properly' classified as "father" and "mother" in the data which

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Morgan gives, while the reciprocal types, MMMBs and MMMBd, which ought to be classified reciprocally as "child" (they are, after all, the children of "uncles"), are classified instead as "grandfather" and "grandmother" on the basis of the conflicting generational principle. Now, since field workers anchor their interrogations to concrete living or at least remembered individuals, the classification of older people is typically obtained from younger people, the classification of younger people typically from older people. Thus, when changes are taking place in tribe's usage, it is the classification of lower-generation types, obtained from older informants, which is the more conservative. This suggests that the generational extensions of these terms may have been an innovation in the system of the Republican Pawnee at the time that the data were collected. The system may have been undergoing change. Hence the conflicting principles of classification.

7.10. Still unresolved are the cyclical identifications which had to be posited in §§7.1–5, all of which may be summarized as $A^{n+3}U = A^nU = UA^{n-3}$, where n is a positive number or zero. Morgan made special note of this curious cycling: "father"—"grandfather"—"uncle"—"father" etc. in the ascending male line, and "son"—"grandson"—"nephew"—"son" etc. in the descending male line. He added further, 'This series must be explained as a refinement upon the common form, designed to discriminate the several ancestors above grandfather and the several descendants below grandson from each other. It is repeated both in the lineal and collateral lines as far as you choose to follow the chain of consanguinity.'[35]

Morgan's explanation of the cycling, as an invention designed to accomplish the discrimination of generation, is of course not adequate. For while it does differentiate generations in the male line, it does so only by sacrificing other discriminations. Rather than consider it a mere device, we must regard it as a natural generalizing response. If the Pawnee classify the FFF and FFFB types as "uncles", we must assume that there is some shared feature of similarity between these and the MB type. Now this 'similarity' is hardly something which inheres in the types themselves and is there for all to see. Rather, it must lodge in some peculiarity of behavior of the Republican Pawnee. We can only speculate as to what this may have been. The ethnographic records do not contain the answer.

There are only two practices known from anthropological literature, which can give rise to a unilineal three-generation cycle. Both are rules governing marriages. One is the circulation of spouses among three sibs, such as is described for the Gilyak.[36] The other is prescribed marriage to a second cross cousin. The first of these is not a possible explanation of the Pawnee cycling, for all reports agree that the Pawnee had no sibs. By elimination, then, we are left

[35] *Systems* 196–7.

[36] Lev Šternberg, *Semja i rod u narodov severo-vostočnoi Azii* (Leningrad, 1933); Josef Haekel, Das Dreiclansystem, *Sitzungsberichte der anthropologischen Gesellschaft in Wien*, Jahrgang 1937–38.11–3 (issued with *Mitteilungen d. A. G. in Wien*, Vol. 68; 1938); Alexander Slawik, Zum Problem des Bärenfestes der Ainu und Giljaken, *Wiener Beiträge zur Kulturgeschichte und Linguistik* 9.189–203 (1952).

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with the second. Such a marriage rule is known elsewhere in the ethnology of the southern United States area. It is documented for the Cherokee, among whom a preferred marriage was of a man to his FFSdd or classificatory equivalent, or, as they have stated it themselves, a man marries his "grandmother".[37]

Let us imagine a hypothetical society in which every man marries his father's father's sister's daughter's daughter. This may strike one at first as a whimsical restriction to impose on the males of a society. But such rules of marriage to specified types of consanguineal kinsmen are not uncommon in primitive societies. They have, moreover, certain definite functions which are their reason for existence. They guarantee replacement of personnel lost to the group through marriage. Loss of personnel is seldom lightly regarded in small social groups living a precarious existence. A hunter and defender is highly valued. So also is a gatherer of wild food and bearer of children. The ever-present incest taboo, however, renders inevitable the loss of the one or the other of these upon marriage—of a male or of a female, depending upon the patterns of residence. Upon such loss to a small unilineal kin group (extended family or lineage), the return in kind may be too important to be left to the workings of chance. It sometimes becomes subject to regulation. With first-cross-cousin marriage (to a FSd or MBd), the return can be made in a generation. With second-cross-cousin marriage (where a society's broader definition of incest renders the first impossible), the return is delayed two generations.

Figure 2 illustrates the consequences of the rule W = FFSdd, for three uterine lineages of our hypothetical society. Note the cycling of Ego's agnatic line around these three uterine lineages. The identifications which result are those which we were forced to posit in our analysis of the Pawnee classifications.

Of course it is too much to assume that in a real society such a rule could always be followed to the letter.[38] More leeway is offered, however, by the substitution of parallel collaterals who are classificatory equivalents. These would conform to the spirit of the rule and would have similar consequences in regard to kin-type identifications and cycling. But even with this additional leeway, exceptions and violations would probably still be inevitable. Kinship systems, however, are shaped by dominant patterns of behavior; perfect conformity is never essential. Evidence from other societies and other systems suffices to show that exceptions to such rules are usually treated as if they were in accord with the rule. A certain incidence of violations may be tolerated without collapse of the pattern.

In view of these considerations, I suggest that a pattern of preferential mating of a man with his female patrilateral second cross cousin must have been responsible for the cycling shown in the data presented by Morgan for the Republican Pawnee. This is not to be inferred for all the Pawnee, however: the data given by Morgan for the Grand Pawnee do not show any three-generation

[37] William H. Gilbert Jr., Eastern Cherokee social organization, *Social anthropology of North American tribes* 283–338 (ed. Fred Eggan; Chicago, 1937, 1955); William H. Gilbert, The eastern Cherokees, *Bureau of American Ethnology, Bulletin* 133.169–414 (1943).

[38] It is obvious that if such a rule were consistently in operation, the society could consist of seven and only seven lineages. Perhaps it is not fortuitous that the Cherokee have seven sibs.

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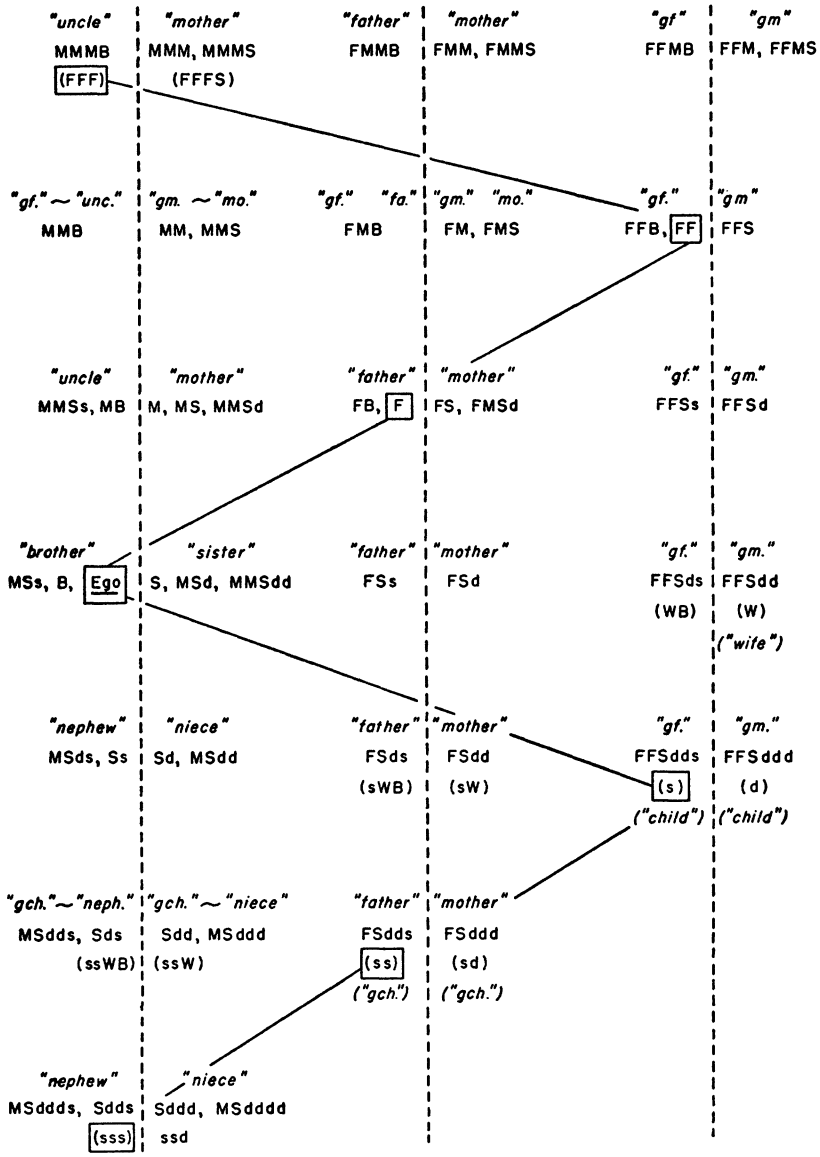


FIGURE 2

Three Uterine Lines: Ego's, his Father's, and his Father's Father's Parenthesized kin types show kin-type identities resulting from patrilateral second-cross-cousin marriage. Boxed kin types are those in Ego's agnatic line. Note the three-generation cycling of the agnatic line among the uterine lines. The kinship terms which stand above the various kin types represent the Pawnee classification of these types.

cycles. Our inference applies at most to the Republican Pawnee, and perhaps only to some of the villages belonging to that group.

In another context (§6) I referred to Kroeber's paper, Classificatory systems of relationship. Here I must refer to it again, for the main import of that paper

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was to caution against the inferring of marriage rules from systems of kinship terminology. Kroeber's attitude toward causal explanations of the semantic structures found in kinship systems was quite similar to the linguist's attitude toward causal explanations of linguistic structures: they are regarded as the product of unique historical linguistic developments whose specific causes are generally not recoverable. The inference of direct sociological causes is regarded as risky, in the absence of historical documentation. At best, according to Kroeber, these structures relate to psychology rather than to sociology, i.e. to cultural attitudes and habits of observation or attention, whose particular genesis is not to be identified in any simplistic fashion with one or another sociological practice. In particular he warned against the invention of hypothetical marriage rules to explain the mergings found in a kinship system.[39]

Kroeber's approach, already in 1909, heralded the modern 'distinctive feature' approach. He showed how classificatory mergings of kin types may result simply from their sharing of certain distinctive features while they differ by other features which are nondistinctive, or whose distinctiveness is neutralized in a particular context of other features, and which therefore go unattended to. He showed how *inattention to a difference* is as satisfactory an explanation of classificatory mergings as the posited *nonexistence of the difference.* Thus he cautioned against indiscriminate inferring of consanguineal marriages in order to effect identities between merged kin types. Rather, it should suffice to point out features of similarity between them.

In our positing of the marriage rule W = FFSdd, and of the other identities which ensue (FFF = MMMB; sss = Sdds; etc.), we seem to be going counter to Kroeber's injunction. We can only plead special immunity, on the following grounds. (1) Inherent features of similarity have already been relied upon to the maximum extent possible to explain the composition of every kin class. The cyclical mergings, which remain over, can be accounted for only by assuming special identities. (2) The rule which we posit is not without precedent.

We should be clear as to the status of our inference: it is hypothesis, not conclusion. The analysis of the semantic structure of a system enables us to form hypotheses regarding social behavior. These hypotheses should be checked against the data of observation. Unfortunately, the checking is now impossible for the Pawnee. We may raise the question and suggest an answer, but there is little likelihood of confirming it. Morgan's data were collected by B. F. Lushbaugh in 1862. Relevant ethnographic field work dates from 1903–7 (Dorsey and Murie) and from 1930–31 (Lesser). Intervening between the first date and the latter dates is the culmination of disaster and changes in Pawnee life begun early in the 19th century. The population was decimated by thirty years of raids and massacres at the hands of the Sioux, which ended only in 1875; by a series of smallpox and cholera epidemics introduced by contact with the whites;

[39] The various discussions provoked by Kroeber's paper form a literature relevant to the 'language and culture' problem. For a review of this, and another anthropologist's opinions, see A. R. Radcliffe-Brown, The study of kinship systems, *Journal of the Royal Anthropological Institute* 71.2 ff. (1941), reprinted as chapter 3 of *Structure and function in primitive society* (Glencoe, Ill., 1952), esp. 59 ff.

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and by periods of starvation due to loss of lands and food supply. Social changes resulted from these causes, from a series of removals and mixing of separate groups, and from determined intervention by military and civil arms of the government and by missionaries, and of course, in more recent times, from submersion in the flood of white society.

8. SOCIOLOGY. We may turn now from speculative sociology to that which is documented.[40]

The Pawnee tribe during the historical period consisted of four bands: the Skiri, the Pitahawirat, the Kitkahaxki, and the Tsawi. Morgan's 'Republican Pawnee', located then on the Republican River near the present boundary between Kansas and Nebraska, were Kitkahaxki. His 'Grand Pawnee' were Tsawi. The 'tribe' was not a political unit, but a linguistic and cultural one, resulting from common origins.

The bands were political units, each being a loose federation of independent villages. The chiefs of the villages formed the band council. The number of villages in a band probably ranged approximately from ten to fifteen. The Skiri were reported at one time to have fifteen villages, but later were reduced to thirteen. The other bands may have been comparable or somewhat smaller. The band, although it was a political unit, can hardly be considered a true social unit. Social life was confined to a smaller unit, the village.

Villages were for the most part independent of each other, except for their rather loose political affiliations into bands. The village was the society; it was the unit within which most social interaction took place. It also constituted the breeding group, being essentially endogamous. Marriages outside the village were rare. The average size of the villages can be only roughly estimated. Village populations probably ranged from one hundred to perhaps three hundred persons. The total population of the Pawnee in the early 19th century has been variously estimated at 4000 to 12,000.

Every village had its sacred bundle, which was its supernatural charter and a focus of religious rites. Each village had a priest and a chief, the rulers respectively over sacred and secular matters. These statuses were passed on in the male line. Class distinctions were marked, in spite of the small scale of these village societies. The basic dichotomy was between nobility and commoners. The nobility consisted for the most part of the families and agnatic kinsmen of priests, chiefs, and medicine men, together with warriors and those wealthy in horses, material possessions, and prestige. Nobility was for the most part a supernatural attribute, hereditary in the male line. Within the warrior group, however, where status could be achieved, there was apparently some room for upward mobility. The noble class seems to have accounted for a little over half of the population in many villages. Commoners were those poor in worldly

[40] George A. Dorsey and James R. Murie, *Notes on Skidi Pawnee society* (Field Museum of Natural History, Anthropological Series, Vol. 27, No. 2; Chicago, 1940); Alexander Lesser, Levirate and fraternal polyandry among the Pawnees, *Man* 30.98–101 (1930); id., *The Pawnee ghost dance hand game* (Columbia University Contributions to Anthropology, Vol. 16; N. Y., 1933). For additional bibliography on the Pawnee, see G. P. Murdock, *Ethnographic bibliography of North America*[2] 80–1 and 198 (New Haven, 1953).

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possessions, influence, and supernatural power. At the extreme fringe of this class were some near outcasts. We have no way of knowing to what extent the classes may have tended toward endogamy. It is not necessary, however, to project onto them the endogamous character of European classes, for even class and caste exogamy are known in North America.[41]

Within the villages the constituent social units were extended families. Each of these shared a large igloo-shaped earth lodge. While rank was inherited in the male line, the family structure and kinship groupings were based on connections in the female line. Potential members of the household were a man, his wife or wives (if more than one, they would be sisters), the wife's parents, the wife's younger unmarried sisters, one of his own brothers who had married one of his wife's sisters, and his (and his brother's) daughters and preadolescent sons. Not every household contained all of these. Some did not include the wife's parents; they might be living with another married daughter and her husband. Some did not have the families of brothers living together; this happened only when brothers married sisters. Some were not polygynous, while some, on the other hand, included as many as four or five wives. The size of the Pawnee dwelling, which usually accommodated ten or twelve beds (laid end to end in two arcs along the circular walls on either side of the lodge), placed limitations on the size of the household. If a potential extended family was too large for a single household, a new lodge would be built, often adjacent, to house the growing family of the son-in-law.

A Pawnee boy, on reaching adolescence, found his future sexual life pretty well mapped out for him in advance. Having entered adolescence, he could no longer live in his boyhood household with his sisters. Modesty and the dread of the thought of incest required his removal. He went to live with his mother's brother, in the household into which this uncle had married. There he became, for a period of several years, a temporary co-husband, along with his uncle, to the latter's wife. It was with his mother's brother's wife that he had his first sexual experience. She initiated him into young manhood. In case there was no mother's brother to take him in, he went instead to live with a married elder brother under similar conditions.

Upon attaining adult competence in warfare, capturing ponies, hunting the buffalo, and providing for a family, the young man married. Marriage was a serious matter, a concern not just of the future husband and wife, but of their families. Marriages were arranged by the families, and in particular, by the respective mothers' brothers of the groom and of the bride. This was especially true among the upper-class Pawnee. The groom and the bride had little or no choice in the matter. If a young man was determinedly opposed to the arrangement being made for him, often his only recourse was flight into temporary exile from his village. Marriages were, in the great majority of cases, to persons of one's own native village. In the rare cases of marriage to someone from another village, the general rule of matrilocality required the husband to move to the

[41] The famous case is that of the Natchez. Cf. John R. Swanton, *Indian tribes of the lower Mississippi Valley and adjacent coast of the Gulf of Mexico* 100–8 (Smithsonian Institution, Bureau of American Ethnology, Bulletin 43; Washington, 1911); C. W. M. Hart, A reconsideration of Natchez social structure, *American anthropologist* 45.374–86 (1943).

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village of his wife. In the still rarer cases, however, where a chief or a priest married someone from a different village, the agnatic succession of these offices took precedence over the general matrilocal rule and required the wife to move to the husband's village.

Upon marriage to a woman, a man automatically acquired the right to all of this woman's younger sisters. As they came of age, he could, if he chose, take them as additional wives in a polygynous household. Or he could make them available to others. Polygyny, when practiced, was always sororal. Successive wives were sisters. Sexual access of brothers to each others' wives was socially approved, but always required consent of the particular brother. As late as World War I, this proved to be something of a convenience. It is said that there were few 'war widows' among the Pawnee; a younger brother simply took over the soldier's wife for the duration. The junior levirate was practiced also. Upon the death of a husband, his younger brother was required to take the widow to wife. In harmony with all of these patterns, and in addition to them, there was some tendency toward marriages of men who were brothers to women who were sisters.

Data on the factors conditioning the choice of a spouse are scant, seemingly contradictory, and largely ambiguous. Even the definition of the incest taboo is poorly recorded. We can be quite certain, however, that the incest taboo extended to all of Ego's recognized parallel kin, uterine and agnatic, regardless of degree of collaterality, and to his father's uterine and agnatic kin also. This eliminates siblings and parallel cousins, and all first cross cousins and their classificatory equivalents, as possible spouses. The way must have been open, however, to second cross cousins and more distant nonparallel consanguineal types. Marriages to distant consanguineals within this accessible sector of one's kindred must have been inevitable in village societies of small size.

A society's marriage rules consist in the drawing of two boundary lines, sharply in some societies, vaguely in others. The first expresses the society's principle of exogamy. This is differently defined in different societies, but in all cases it describes an inner group within which one *should not* marry. It is the society's incest taboo. The second expresses the society's principle of endogamy, of preferential matings. This also is variously defined, depending on the society. But however defined (by class, caste, race, religion, consanguineal kin group, etc.), it describes a group within which one *should* marry. In some societies, such as our own, the outer and inner boundaries are far apart, leaving between them an indefinitely large range of choice for selecting a spouse. In other societies, especially in those with consanguineal marriage prescriptions, the boundaries may be so close together as to leave very little opportunity for choice.

We have already noted the prevailing village endogamy among the Pawnee. Beyond this we have only one statement in the ethnographic literature hinting at a further narrowing of the marriageable group: 'Pawnee marriage customs did involve *the mating of kin at a certain distance removed.* In their old life, families who were the proud possessors of sacred and esoteric lore attempted to retain control of these things by regulating intermarriage.'[42] Could 'kin at a certain distance

[42] Lesser, *The Pawnee ghost dance hand game* 42.

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removed' refer to the FFSdd? One may note that under the above incest rules, the closest marriageable kin of Ego's generation would be just this. This statement is the only one in the literature which may lend, if not support, at least plausibility to the inference drawn in §7.10. Outside of this there is just the compelling nature of the kinship data.

With this brief sketch of Pawnee sociology, their kinship classifications may now take on a certain aspect of reasonableness. It is quite proper that a man's mother's brother's wife should be "wife", and so also his brother's wife and his wife's sister. Also there is significance in the fact that kinship terms which are applied to members of one's nuclear family are extended to all persons who fall under the incest taboo. Parallel cousins, no matter how many degrees of collaterality removed, are "brothers" and "sisters". First cross cousins are "father" and "mother" on one side and "son/daughter" ("child") on the other side. One must not marry a primary kinsman, nor a classificatory equivalent to one of these. The social equivalence and substitutability of brothers which is found in the social structure is fully reflected in the structure of the terminological paradigm. The position of the mother's brother as a sort of special elder brother, and of the sister's son as a special younger brother, are made clear in social behavior. These also are reflected in the terminological system, if not directly in the terms for the maternal uncle and the sororal nephew, at least indirectly in the fact that these participate in the 'equivalence of brothers' when it comes to the classification of their offspring: an "uncle's" child, a "nephew's" child, a "brother's" child, and a man's own child, are all "child" to him. The equivalence of brothers is a special case of a more general social equivalence of uterine kinsmen of one sex and of their substitutability in certain contexts. This relates to the overriding of generations in uterine lines in a large part of the system. Even the peculiar measure of kinship distance that we called agnatic rank, which is basic to the system, and in whose reckoning generation distance is measured up and down male lines but is counted as zero up and down female lines, has its special appropriateness. This sort of reckoning is frequently found in societies where kin aggregates and social solidarity units follow lines of uterine connection while some important social attribute follows the male line and renders agnatic descent meaningful.

III. Conclusion

9. Kinds of meaning. There are many facets to the 'meaning' of a word.[43] A familiar dichotomy is that between reference and connotation. More broadly, we may distinguish between situational meanings and behavioral meanings, saying with Bloomfield that the 'meaning' of a linguistic form consists of 'the situation in which the speaker utters it and the response which it calls forth in the hearer'.[44] If we ask, What stimuli elicit a given language form as a response?, we are inquiring into situational meaning. If we ask, What responses are elicited

[43] I have discussed these in more detail in *Psycholinguistics: A survey of theory and research problems* 171–7 (C. E. Osgood and T. A. Sebeok, editors; Indiana University Publications in Anthropology and Linguistics, Memoir 10; 1954).

[44] Bloomfield, *Language* 139.

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by the given language form as a stimulus?, we are inquiring into behavioral meaning. Language forms function dually as responses to antecedent stimuli, and as stimuli to further responses. The antecedent stimuli and the further responses are parts of the meaningful context of a language form. The former relate to 'semantics', the latter to 'pragmatics' as defined by Morris (see §1).

Antecedent stimuli may include linguistic as well as nonlinguistic stimuli. Further responses also may be partly linguistic and partly nonlinguistic. If we separate out the linguistic portions from both sides of the total context (or contexts) of a linguistic form, we have the data of LINGUISTIC MEANING. From this one may arrive at contextual definitions for linguistic forms. Logicians and structural linguists have experimented with the analysis of this kind of meaning. Some linguists have attempted to let the analysis of linguistic meaning substitute for that of 'real' or 'outside' meaning.[45] The former is said to be within the province of the linguist, while the latter is left to the anthropologist and the psychologist. Linguistic meaning belongs to Morris's 'syntactics'. The linguistic meaning of a form consists of its relations, syntagmatic and paradigmatic, to other forms in the language.

In §§4–7 our concern was exclusively with referential meanings, i.e. with semantics. Referential meaning is of primary concern to the lexicographer. The anthropologist and the psychologist, on the other hand, are interested mainly in behavioral meaning, or pragmatics. An anthropologist uses semantic analysis of a society's kinship terminology only as a tool for getting at the structure of nonlinguistic behavior within the family and kindred in that society. The best works in this field have consisted of careful semantic studies followed by careful studies of social interaction between variously related pairs of kinsmen. The system of discriminatory linguistic behavior is then related to the system of discriminatory nonlinguistic behavior.[46]

Our analysis here has been of semantic structure. We have touched only briefly and unsystematically on pragmatics, but we have seen how semantic analysis may be used to raise questions and to give direction to the observation of nonlinguistic behavior. Pragmatics also is susceptible of structural analysis. The trilogy would be completed by linguistic analysis. This would be concerned not just with the inner structure or morphology of the forms (which would be largely irrelevant), but in particular with their extended relations with other forms. To the best of my knowledge, no one has yet attempted to study these for a set of kinship terms in any language.

A complete study of a kinship system would consist of analyses of these three kinds of meaning—referential, behavioral, and linguistic—and a relating of the three structures one to another. We should not be surprised if the three structures turn out to be isomorphic, or disturbed if they do not. The relating of these three kinds of structure is the essence of the 'language and culture' problem.

[45] Cf. Martin Joos, The description of language design, *Journal of the Acoustical Society of America* 22.701–8 (1950), esp. 708.

[46] As examples of such studies see Ward H. Goodenough, *Property, kin, and community on Truk* (Yale University Publications in Anthropology, No. 46; New Haven, 1951), and the papers by various authors in Fred Eggan (ed.), *Social anthropology of North American tribes* (Chicago, 1937, 1955).

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10. 'SAME OR DIFFERENT'. The reader may be surprised to find here no analysis of the Pawnee linguistic forms. He may have noted similarities between forms, such as might be indicative of common prefixes in some cases, or of a common root in other cases. A linguist's habitual curiosity in regard to such matters may cause him to be left unsatisfied if he finds no discussion of these things. But there will be none.[47] We shall defend the position that, for semantic analysis, they are essentially irrelevant.

This is a position which most American structural linguists should find it embarrassing to attack; for they habitually maintain the converse: that for purposes of linguistic analysis, the analysis of semantics is irrelevant. They insist that for linguistic analysis we need inquire no further into meanings than is necessary to obtain an answer to the question: Are these meanings the 'same' or are they 'different'? All we depend upon is an informant's response to this question. It is thus that one distinguishes between 'free variation' and 'contrast'. If a linguist who knows no English records from my lips the word *cat* first with a final aspirated stop and later with a final preglottalized unreleased stop, the phonetic data will not tell him whether these forms contrast or not. It is only when he asks me, his informant, whether the meaning of the first form is different from that of the second, and I say it is not, that he will be able to proceed with his phonemic analysis.[48] Anything else which I volunteer to tell him about cats and what they mean to me, however, will be irrelevant and may prompt him to hush me so that he can get on with his work.

There is a justification for both of these positions: for the independence of linguistics from semantics, and for the independence of semantics from linguistics. We may argue it as follows.

All responses of an organism are classificatory: they effect classifications of stimuli. An organism does not respond differently to every objectively different stimulus situation. A situation contains an indefinitely large number of features, any of which is potentially significant. How an individual responds depends upon what features he has learned to perceive, to be on the alert for, and to react to in a consistent manner. Features which are significant to one individual, and which serve as cues for responses, may be disregarded by another. Thus, the same situation may be differently classified by the responses of two different individuals.

Language forms, because of their dual role as response and as stimulus, become on the one hand classifiers (see Figure 3*a*) and on the other hand classified (Figure 3*b*).

A simple illustration of language forms as classifiers is in the naming response. Not every objectively different stimulus object or stimulus attribute in the environment has a different name. Classifications are made. But the mode of classification and the location of class boundaries often vary greatly from one society to

[47] Works on Pawnee linguistics which may be consulted: J. Dunbar, *The Pawnee language* (appendix to G. B. Grinnell, *Pawnee hero stories and folk-tales*[2] 409–37; New York, 1893); Gene Weltfish, The vision story of Fox-Boy, a South Band Pawnee text, *International journal of American linguistics* 9.44–75 (1936); id., *Caddoan texts* (Publications of the American Ethnological Society, Vol. 17; N. Y. 1937).

[48] Cf. Bloomfield, *Language* 76–8.

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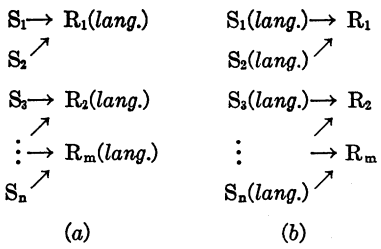


FIGURE 3

(*a*) Language forms as classifiers of situations.

(*b*) Language forms as classified by further responses.

S(*lang.*) and R(*lang.*) are language forms in their two roles as stimuli and as responses. S and R are meanings, semantic and pragmatic.

another. Color terms, ethnobotanical names, and kinship terminology are among the examples most often cited in illustration of this point. But the same is true in almost all areas of vocabulary. We have demonstrated it for a kinship terminology in §§4–7.

A simple illustration of language forms in the role of the classified may be taken from phonemics. The difference between two objectively different phonetic stimuli is either utilized as a cue for differentiating responses by the members of a society, or it is not utilized. In the first case the stimuli get classified separately, as 'different'. In the latter case they get thrown together as the 'same', or as stimulus equivalents. Here also the mode of classification and the location of class boundaries vary greatly from one society to another.

It is this 'same or different' which is essential. In linguistic analysis we define contrast among forms operationally in terms of difference in meaning responses. In semantic analysis we define contrast among meanings operationally on the basis of difference in language responses. The fact that many astute students of kinship semantics have been astoundingly poor linguists has not been a serious handicap to them. At least they can usually hear whether two kinship terms (linguistic forms) are the same or different, and if they have any doubt they can get an unambiguous answer from their informants. Whether their transcriptions are elegant or atrocious is of little import in this context. (Morgan's transcriptions are among the worst.) All that is necessary is that different terms be somehow orthographically distinguished. Beyond this minimal requirement, students of kinship are often as cavalier in their treatment of linguistic forms as linguists tend to be in their treatment of meanings.

There are certain parallels between the analytic constructs of semantics and those of linguistics which are worth noting. They need not be regarded either as fortuitous or as forced analogizing. Rather, they follow from similar bases for classification, namely, from a society's learned habits of discriminatory responses. Compare the following from phonology and from kinship semantics.

Phones. Unique events.	*Kinsmen.* Unique individuals.
Phone type. A class of phones heard and transcribed as the same by the	*Kin type.* A class of kinsmen given the same designation by the eth-

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phonetician. [The phone types are a product of cross-cultural sophistication on the part of the investigator. What he hears as the same and what he hears as different depend on his perception habits. These depend on his previous training, and represent a composite of the phonetic discriminations made in many languages. The discriminations are sufficiently fine so as to reduce the probability of the investigator's obscuring a phonetic difference which may be distinctive in the language under study.]

Phoneme. A class of noncontrastive phone types which share a distinctive bundle of phonetic features. [A phoneme can be defined by listing the principal phone types (allophones) which belong to it; or it can be defined by stating the defining phonetic features of the class.]

nologist. [The kin types with which the ethnologist works are a product of cross-cultural sophistication. They represent a composite of the semantic discriminations made in many societies. They are sufficiently finely drawn so as to reduce the probability of the investigator's obscuring a semantic difference which may be distinctive in the society under study.]

Kin class. A class of kin types which are not contrasted terminologically and which share some distinctive bundle of semantic features. [A kin class can be defined by listing the principal kin types which belong to it; or it can be defined by stating the defining semantic features of the class.]

One might continue with further parallels in the componential (paradigmatic) structure of the systems, in the combinatorial (syntagmatic) patterns, etc.

We have attempted to show, through the example of a kinship system, that the analysis of semantic structure is an operation separate from and not dependent upon the analysis of linguistic structure. Kinship terminology is but a special area of vocabulary, and its structure is a special case of semantic structure. Our 'kinsman', 'kin type', and 'kin class' are thus special cases of more general analytic constructs which are not yet named. The terms SEME, SEME TYPE, and SEMEME may be more fitting in this use than in their previous application as more or less superfluous labels for linguistically defined units.[49]

11. METAPHOR. If denotata, definable componentially as $A{\cdot}B{\cdot}X$, $A{\cdot}B{\cdot}Y$, $A{\cdot}B{\cdot}Z$, etc., constitute a class, that class can be defined as $A{\cdot}B$. The features A and B are called, in various terminologies, the 'defining features', the 'distinctive

[49] As in Bloomfield, *Language* 162; E. A. Nida, A system for the description of semantic elements, *Word* 7.1–14 (1951); and W. L. Wonderly, Semantic components in Kechua person morphemes, *Lg.* 28.366–76 (1952). In the latter two works, semes and sememes are not restricted to the meanings of morphs and morphemes, but may sometimes be identified with semantic components. In our suggested application of the terms, there could be no equating of sememe with semantic component any more than there is of equating phoneme with phonetic component.

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features', or the 'essential features' of the class, or the 'necessary and sufficient conditions' for membership in the class. The features X, Y, Z, etc., on the other hand, are 'nondistinctive features' or 'accidental features' of particular members of the class. Suppose that a society has a name for this class, but suppose now that they begin to apply the same name also to denotata $C \cdot Y$, $M \cdot N \cdot Y$, etc. as well. Now the possession of the feature Y, which was an accidental feature of one of the members of the original class, has become also a sufficient (but not a necessary) condition for membership in the class. As a result, $A \cdot B$ are no longer necessary and sufficient conditions, but merely sufficient conditions. The class Y is a METAPHORIC EXTENSION from the original class $A \cdot B$, which it intersects. The composite class, consisting of the original one together with its metaphoric extension, can no longer be given a unitary definition. The Pawnee "grandfather" and "grandmother" classes are examples of such composite classes, with their extensions based on a generation feature, G^2, which is a mark of some but not all of the members of the basic class. We have also spoken of the Pawnee reference to corn as "Our Mother", a metaphoric extension of the "mother" term.

The putting out of metaphoric extensions from an original class of meanings, and of extensions to these extensions, constitutes the normal process of semantic drift. An original meaning may in time be lost sight of. This process affects nearly all areas of vocabulary. Most words have designata so complex in metaphoric extension that unitary definitions in terms of simple class products are impossible. This does not, however, necessarily render our method of analysis inapplicable. There are many areas of vocabulary which can be analysed by this method if semantic fields are first carefully delineated. Such a field should be covered by a paradigm of terms whose meanings have, *within this delimited area*, a basic semantic feature in common, but whose meanings contrast with each other by virtue of one or more differences in respect to several other kinds of features. Metaphoric extensions outside the delimited area, i.e. denotata which lack the basic feature common to the set, are for the time being disregarded. Thus, if we were to set out now to analyse the kinship system of the Iroquois, we should leave out of the picture 'Our Mother the Earth', 'Our Grandmother the Moon', 'Our Elder Brother the Sun', 'Our Grandfathers the Thunderers', and 'The Three Sisters, the Corn, the Beans, and the Squash', because these do not fall within the semantic field determined by an objectively defined common semantic component K 'kinsman'. We would bring in these personages later, after the basic field had been analyzed. We would treat them as metaphoric extensions, and would seek to discover the 'accidental' features which serve as bases for these extensions and which suggested them to the Iroquois.

In some areas of lexicon, semantic structure may be so complex that it is impossible or unprofitable to approach it in this manner, with Aristotelian class logic and the 'same or different' pragmatic test as the principal tools. It may become necessary to abandon the Aristotelian dichotomy of A vs. *not-A*, and the clear distinction between 'essential' and 'accidental' features. Continuous scales may be introduced in place of these sharp dichotomies; degrees of 'criteriality'

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may be measured for various semantic features; and tests other than the simple 'same or different' may have to be devised.[50] Nonetheless our simpler tools and methods have a wide utility in semantic analysis.[51]

[50] I am indebted to Roger W. Brown for these suggestions, which he offered in discussion of an earlier version of this paper at a Linguistic Forum meeting, Linguistic Institute, University of Michigan, in the summer of 1954.

[51] The simpler approach has had a wide utility in phonemics. One may raise the question whether in the future it may be necessary to alter these procedures, especially in relating data on perception of phonemes to data on acoustics.

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